

Name: _____ Date: _____

Period: _____

Exit Ticket — Key

1. The equation $4x^2 + 9y^2 = 36$ describes an ellipse.

(a) For $x = 3$: $4(9) + 9y^2 = 36 \implies 36 + 9y^2 = 36 \implies 9y^2 = 0 \implies y = 0$

$y = 0$ (only one value)

(b) For $x = 0$: $9y^2 = 36 \implies y^2 = 4 \implies y = \pm 2$

$y = \pm 2$

2. Consider moving along the **upper half** of the ellipse $4x^2 + 9y^2 = 36$ from the point $(3, 0)$ toward the point $(0, 2)$.

(a) As x decreases from 3 to 0, does y increase or decrease?

y increases (from 0 to 2).

(b) Is the co-variation between x and y positive or negative? Explain.

Negative. As x decreases, y increases — they change in opposite directions, so $\Delta y / \Delta x < 0$.

Teacher Note

Q1(a): $(3, 0)$ is on the ellipse; at $x = 3$ the ellipse just touches the x -axis (one y -value, not two). Some students may be surprised — this is a good discussion point.

Q2: The key insight is that “opposite directions” means negative co-variation even when x is decreasing. $\Delta x < 0$ and $\Delta y > 0$, so $\Delta y / \Delta x < 0$. Award full credit for any correct explanation referencing opposite directions or the sign of the ratio.